Neurotoxic effects of aluminium among foundry workers and Alzheimer's disease.

BACKGROUND: In a cross-sectional case-control study conducted in northern Italy, 64 former aluminium dust-exposed workers were compared with 32 unexposed controls from other companies matched for age, professional training, economic status, educational and clinical features. The findings lead the authors to suggest a possible role of the inhalation of aluminium dust in pre-clinical mild cognitive disorder which might prelude Alzheimer's disease (AD) or AD-like neurological deterioration. METHODS: The investigation involved a standardised occupational and medical history with particular attention to exposure and symptoms, assessments of neurotoxic metals in serum: aluminium (Al-s), copper (Cu-s) and zinc (Zn-s), and in blood: manganese (Mn-b), lead (Pb-b) and iron (Fe-b). Cognitive functions were assessed by the Mini Mental State Examination (MMSE), the Clock Drawing Test (CDT) and auditory evoked Event-Related Potential (ERP-P300). To detect early signs of mild cognitive impairment (MCI), the time required to solve the MMSE (MMSE-time) and CDT (CDT-time) was also measured. RESULTS: Significantly higher internal doses of Al-s and Fe-b were found in the ex-employees compared to the control group. The neuropsychological tests showed a significant difference in the latency of P300, MMSE score, MMSE-time, CDT score and CDT-time between the exposed and the control population. P300 latency was found to correlate positively with Al-s and MMSE-time. Al-s has significant effects on all tests: a negative relationship was observed between internal Al concentrations, MMSE score and CDT score; a positive relationship was found between internal Al concentrations, MMSE-time and CDT-time. All the potential confounders such as age, height, weight, blood pressure, schooling years, alcohol, coffee consumption and smoking habit were taken into account. CONCLUSIONS: These findings suggest a role of aluminium in early neurotoxic effects that can be detected at a pre-clinical stage by P300, MMSE, MMSE-time, CDT-time and CDT score, considering a 10 micrograms/l cut-off level of serum aluminium, in aluminium foundry workers with concomitant high blood levels of iron. The authors raise the question whether pre-clinical detection of aluminium neurotoxicity and consequent early treatment might help to prevent or retard the onset

of AD or AD-like pathologies.